实验 B 文档数据库 MongoDB 实验

实验说明：

1. 实验目标

通过本实验掌握 MongoDB 基本概念，以及利用 MongoDB 对文档进行增删改查等基本操作，同时能完成简单的 MongoDB 链接操作及地理位置索引操作。

2. 实验环境：Window 系统安装 MongoDB v4.0.6。

实验任务：

请完成下面三个实例题目，并将相应代码截图附于每小题后面。

任务一：创建名为 XN 的数据库，在该数据库下创建集合 dsj,并在 dsj 集合中插入下列文档:
{item:"journal",qty:25,size:{h:14,w:21,uom:"cm"},status:"A"},
{item:"notebook",qty:50,size:{h:8.5,w:11,uom:"in"},status:"A"},
{item:"paper",qty:100,size:{h:8.5,w:11,uom:"in"},status:"D"},
{item:"planner",qty:75,size:{h:22.85,w:30,uom:"cm"},status:"D"},
{item:"postcard",qty:45,size:{h:10,w:15.25,uom:"cm"},status:"A"},
{item:"paper",qty:45,size:{h:20,w:25,uom:"cm"},status:"A"}
针对上述集合，执行相关命令分别实现下述要求(所有步骤需提供相应命令及执行结果截图):

```
> db.XN.dsj.insertMany([{item:"journal",qty:25,size:{h:14,w:21,uom:"cm"},status:"A"},{item:"notebook",qty:50,size:{h:8.
,w:11,uom:"in"},status:"A"},{item:"paper",qty:100,size:{h:8.5,w:11,uom:"in"},status:"D"},{item:"planner",qty:75,size:{h
22.85,w:30,uom:"cm"},status:"D"},{item:"postcard",qty:45,size:{h:10,w:15.25,uom:"cm"},status:"A"},{item:"paper",qty:45,
ize:{h:20,w:25,uom:"cm"},status:"A"}])
{
        "acknowledged" : true,
        "insertedIds" : [
                ObjectId("654c44480e48376a3e176c7f"),
                ObjectId("654c44480e48376a3e176c80"),
                ObjectId("654c44480e48376a3e176c81"),
                ObjectId("654c44480e48376a3e176c82"),
                ObjectId("654c44480e48376a3e176c83"),
                ObjectId("654c44480e48376a3e176c84")
        ]
}
```

1) 对 dsj 集合下所有文档按照 qty 值升序排序显示，并要求只显示前 3 个文档。

```
> db.dsj.find().sort({qty:1}).limit(3)
{ "_id" : ObjectId("654c421d0e48376a3e176c79"), "item" : "journal", "qty" : 25, "size" : { "h" : 14, "w" : 21, "uom" : "
cm" }, "status" : "A" }
{ "_id" : ObjectId("654c421d0e48376a3e176c7d"), "item" : "postcard", "qty" : 45, "size" : { "h" : 10, "w" : 15.25, "uom"
 : "cm" }, "status" : "A" }
{ "_id" : ObjectId("654c421d0e48376a3e176c7e"), "item" : "paper", "qty" : 45, "size" : { "h" : 20, "w" : 25, "uom" : "cm
" }, "status" : "A" }
>
```

2) 筛选满足条件（status:"A"）的首个文档，同时该文档要求只显示 item 及 qty 内容。

```
> db.dsj.find({status:"A"},{_id:0,item:1,qty:1})
{ "item" : "journal", "qty" : 25 }
{ "item" : "notebook", "qty" : 50 }
{ "item" : "postcard", "qty" : 45 }
{ "item" : "paper", "qty" : 45 }
>
```

3) 筛选 25≤qty≤80 的文档，并统计复合条件的文档数量。

```
> db.dsj.find({qty:{$gte:25,$lte:80}}).count()
5
```

4) 筛选满足条件(item:"paper")的第一个文档,并修改 qty 的值为 0,同时增设 lastModified 字段以记录最近的修改日期。最后查询显示所有文档信息。

```
> db.dsj.update({item:"paper"},{$set:{"qty":0},$currentDate:{lastModified:true}})
WriteResult({ "nMatched" : 1, "nUpserted" : 0, "nModified" : 1 })
> db.dsj.find()
{ "_id" : ObjectId("654c421d0e48376a3e176c79"), "item" : "journal", "qty" : 25, "size" : { "h" : 14, "w" : 21, "uom" : "
cm" }, "status" : "A" }
{ "_id" : ObjectId("654c421d0e48376a3e176c7a"), "item" : "notebook", "qty" : 50, "size" : { "h" : 8.5, "w" : 11, "uom" :
 "in" }, "status" : "A" }
{ "_id" : ObjectId("654c421d0e48376a3e176c7b"), "item" : "paper", "qty" : 0, "size" : { "h" : 8.5, "w" : 11, "uom" : "in
" }, "status" : "D", "lastModified" : ISODate("2023-11-09T02:46:05.813Z") }
{ "_id" : ObjectId("654c421d0e48376a3e176c7c"), "item" : "planner", "qty" : 75, "size" : { "h" : 22.85, "w" : 30, "uom"
: "cm" }, "status" : "D" }
{ "_id" : ObjectId("654c421d0e48376a3e176c7d"), "item" : "postcard", "qty" : 45, "size" : { "h" : 10, "w" : 15.25, "uom"
 : "cm" }, "status" : "A" }
{ "_id" : ObjectId("654c421d0e48376a3e176c7e"), "item" : "paper", "qty" : 45, "size" : { "h" : 20, "w" : 25, "uom" : "cm
" }, "status" : "A" }
>
```

5) 筛选并更新满足条件（item:"paper"）的所有文档， 增设字段 Marked：XXXY。

```
> db.dsj.update({item:"paper"},{$set:{"Marked":"XXXY"}})
WriteResult({ "nMatched" : 1, "nUpserted" : 0, "nModified" : 1 })
> db.dsj.find()
{ "_id" : ObjectId("654c421d0e48376a3e176c79"), "item" : "journal", "qty" : 25, "size" : { "h" : 14, "w" : 21, "uom" : "
cm" }, "status" : "A" }
{ "_id" : ObjectId("654c421d0e48376a3e176c7a"), "item" : "notebook", "qty" : 50, "size" : { "h" : 8.5, "w" : 11, "uom" :
 "in" }, "status" : "A" }
{ "_id" : ObjectId("654c421d0e48376a3e176c7b"), "item" : "paper", "qty" : 0, "size" : { "h" : 8.5, "w" : 11, "uom" : "in
" }, "status" : "D", "lastModified" : ISODate("2023-11-09T02:46:05.813Z"), "Marked" : "XXXY" }
{ "_id" : ObjectId("654c421d0e48376a3e176c7c"), "item" : "planner", "qty" : 75, "size" : { "h" : 22.85, "w" : 30, "uom"
: "cm" }, "status" : "D" }
{ "_id" : ObjectId("654c421d0e48376a3e176c7d"), "item" : "postcard", "qty" : 45, "size" : { "h" : 10, "w" : 15.25, "uom"
 : "cm" }, "status" : "A" }
{ "_id" : ObjectId("654c421d0e48376a3e176c7e"), "item" : "paper", "qty" : 45, "size" : { "h" : 20, "w" : 25, "uom" : "cm
" }, "status" : "A" }
>
```

任务二：对于下述两个集合:

```
db.A.insertMany ([
    { "_id" : 1, "sku" : "almonds", description: "product 1", "instock" : 120 },
    { "_id" : 2, "sku" : "bread", description: "product 2", "instock" : 80 },
    { "_id" : 3, "sku" : "cashews", description: "product 3", "instock" : 60 },
    { "_id" : 4, "sku" : "pecans", description: "product 4", "instock" : 70 },
    { "_id" : 5, "sku": null, description: "Incomplete" },
    { "_id" : 6 }
])

db.B.insertMany([
    { "_id" : 1, "item" : "almonds", "price" : 12, "quantity" : 2 },
    { "_id" : 2, "item" : "pecans", "price" : 20, "quantity" : 1 },
    { "_id" : 3   }
])
```

请执行相关命令，使得集合 B 中文档可以关联显示集合 A 中的文档。(要求提供相应命令及执行结果截图)。

```
db.B.aggregate([
     {
          $lookup:
          {
            from:"A",
```

localField:"item",
foreignField:"sku",
as: "res_docs"
}}])

```
> db.B.aggregate([{ $lookup: { from:"A", localField:"item", foreignField:"sku", as: "res_docs" }}])
{ "_id" : 1, "item" : "almonds", "price" : 12, "quantity" : 2, "res_docs" : [ { "_id" : 1, "sku" : "almonds", "descripti
on" : "product 1", "instock" : 120 } ] }
{ "_id" : 2, "item" : "pecans", "price" : 20, "quantity" : 1, "res_docs" : [ { "_id" : 4, "sku" : "pecans", "descriptio
" : "product 4", "instock" : 70 } ] }
{ "_id" : 3, "res_docs" : [ { "_id" : 5, "sku" : null, "description" : "Incomplete" }, { "_id" : 6 } ] }
```

任务三：（文件夹：04JC9-JSON 格式数据文件）实例提供两个存储有一些饭店位置信息的 JSON 集合数据（restaurants.josn, neighborhoods.josn），现假设用户移动设备定位当前所在位置的经纬度信息为[-73.93414657,40.82302903]，请执行相关命令实现下述查询要求：

1) 对照 neighborhoods.josn 标定区域，查询用户所属区域，并根据 restaurants.josn 统计当前所在区域中的饭店数量。

```
db.nei.findOne({
    geometry:
    {
        $geoIntersects:
        {
            $geometry:
            {
                type:'Point',
                coordinates:[ -73.93414657,40.82302903]
             }
        }
    }
})
```

Raw shell output | Shell Output (Documents)

50 | Documents 1 to 1

Result > name

_id	geometry	name
55cb9c666c52...	{ 2 fields }	Central Harlem North-Polo Grounds

```
            {
                type:'Point',
                coordinates:[ -73.93414657,40.82302903]
            }
        }
    }
})

db.res.find({
    location:
    {
        $geoWithin:
        {
            $geometry:neb.geometry
        }
    }
}).count()
```

Raw shell output

Restart the MongoDB Shell | Use legacy shell | Clear raw shell output | Pin r

```
//   - Run Shell Queries via Studio 3T for full access to query resul
// ----------------------------------------------------------------

Current Mongosh Log ID: 654cd8c45079aa130973366a
Using Mongosh:      1.8.0

For mongosh info see: https://docs.mongodb.com/mongodb-shell/

>
> admin
switched to db MM

127
```

2) 查询离用户 2 英里以内的饭店数量。

```
db.res.find({
    location:
    {
        $geoWithin:
        {
            $centerSphere:[[-73.93414657,40.82302903],2]
        }
    }
}).count()
```

Raw shell output

Restart the MongoDB Shell | Use legacy shell | Clear raw shell output | Pin new results

```
// ------------------------------------------------------------------------

Current Mongosh Log ID: 654cd8c45079aa130973366a
Using Mongosh:      1.8.0

For mongosh info see: https://docs.mongodb.com/mongodb-shell/

>
> admin
switched to db MM

127
25353
```